

Dimension-Free Eldan–Gross Inequalities for Coupled Strongly Log-Concave Measures

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Abstract

We prove a dimension-free Eldan–Gross inequality for arbitrary strongly log-concave measures on Euclidean space, with influences measured in a fixed coordinate system. Let $d\mu = Z^{-1}e^{-V} dx$ on $\mathbb{R}^n$, where $\nabla^2 V \succeq \alpha I$, and let E have finite weighted perimeter. Writing $I_i^\mu(E)$ for its geometric influence in coordinate i and $J_\mu(E) = \sum_i I_i^\mu(E)^2$, we establish

$$P_\mu(E) \geq c\sqrt{\alpha} \mu(E)(1 - \mu(E)) \sqrt{\log\left(1 + \frac{c\alpha}{J_\mu(E)}\right)}$$

with a universal constant $c > 0$. No product structure, upper Hessian bound, conditional independence, or weak-interaction hypothesis is assumed. More generally, for every $f : \mathbb{R}^n \rightarrow [0, 1]$ of finite weighted variation, we prove

$$\mathrm{TV}_\mu(f) \geq c\sqrt{\alpha} \mathrm{Var}_\mu(f) \sqrt{\log\left(1 + \frac{c\alpha}{\sum_i |D_i f|_\mu(\mathbb{R}^n)^2}\right)}.$$

This specializes exactly to the set inequality at $f = \mathbf{1}_E$.

The proof has three ingredients. First, a Helffer–Sjöstrand resolvent comparison gives the smooth L^1 – L^2 Talagrand inequality in the coupled setting. Second, a data-dependent nonlinear Minkowski heat flow regularizes bounded BV data while contracting a crystalline variation and controlling an arbitrary diagonal quadratic energy. Third, a compact-convex minimax argument makes all coordinate energy estimates simultaneous. A moment interpolation then recovers the logarithm. We include the whole-space nonlinear-flow construction, its weak Bochner–Kato estimate through the zero-gradient set, and all weighted- BV limiting steps.

Keywords. Geometric influences; Eldan–Gross inequality; strongly log-concave measure; nonlinear Finsler heat flow; Talagrand inequality; weighted perimeter.

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1 Introduction

Influence inequalities form a central interface between analysis of Boolean functions, threshold phenomena, concentration, and theoretical computer science. Talagrand’s

inequality strengthens Poincaré by charging a coordinate according to the ratio between its L^2 and L^1 derivative norms [16]. Eldan and Gross subsequently proved a sharp isoperimetric consequence on the Boolean cube in which the square root of pointwise sensitivity is bounded below by the variance times a logarithm of the reciprocal squared-influence sum [7]. Later proofs and extensions appear in [8, 11].

For a continuous measure, the natural analogue of coordinate influence is the coordinate projection of the surface measure. If

$$d\mu(x) = p_\mu(x) dx = Z^{-1}e^{-V(x)} dx$$

and $E \subseteq \mathbb{R}^n$ has finite weighted perimeter, define

$$\begin{aligned} P_\mu(E) &= \int_{\partial^* E} p_\mu d\mathcal{H}^{n-1}, \\ I_i^\mu(E) &= \int_{\partial^* E} |\nu_{E,i}| p_\mu d\mathcal{H}^{n-1}, \\ J_\mu(E) &= \sum_{i=1}^n I_i^\mu(E)^2. \end{aligned}$$

Here $\partial^* E$ is the reduced boundary and ν_E its measure-theoretic Euclidean unit normal.

For product measures, coordinatewise gradient contraction lets one transfer Talagrand’s smooth inequality to an indicator at the correct scale. This mechanism fails for coupled measures. Indeed, curvature controls

$$\|\nabla f\|_{L^1(\ell_2)},$$

whereas the Eldan–Gross quantity involves

$$\|\nabla f\|_{\ell_2(L^1)} = \left(\sum_i \|\partial_i f\|_1^2 \right)^{1/2}.$$

These mixed norms can differ by $\sqrt{n}$, and a Witten multiplier need not be bounded in the entrywise-absolute regular norm. Thus the product proof cannot simply be repeated.

Our main result removes the dimension loss without imposing any commuting coordinate structure.

Theorem 1.1 (Set theorem). *There are universal constants $c_0, c_1 > 0$ such that the following holds. Let $n \geq 1$, $\alpha > 0$, and $d\mu = Z^{-1}e^{-V} dx$ on $\mathbb{R}^n$, where $V \in C^2(\mathbb{R}^n)$*

and $\nabla^2 V \succeq \alpha I$. For every Borel set E of finite weighted perimeter,

$$P_\mu(E) \geq c_0 \sqrt{\alpha} \mu(E) (1 - \mu(E)) \sqrt{\log \left(1 + \frac{c_1 \alpha}{J_\mu(E)} \right)}.$$

If $\mu(E) \in \{0, 1\}$, the right-hand side is interpreted as zero.

For a weighted BV function f , write

$$\begin{aligned} \mathrm{TV}_\mu(f) &= \int_{\mathbb{R}^n} p_\mu \, d|Df|, \\ I_i^\mu(f) &= \int_{\mathbb{R}^n} p_\mu \, d|D_i f|, \\ J_\mu(f) &= \sum_{i=1}^n I_i^\mu(f)^2. \end{aligned}$$

For $f \in W^{1,1}(\mu)$, these are respectively $\int |\nabla f| \, d\mu$, $\int |\partial_i f| \, d\mu$, and the squared ℓ_2 -sum of the latter quantities.

Theorem 1.2 (Functional theorem). *There are universal constants $c_2, c_3 > 0$ such that, under the assumptions of Theorem 1.1, every $f \in BV_{\mathrm{loc}}(\mathbb{R}^n)$ with $0 \leq f \leq 1$ and $\mathrm{TV}_\mu(f) < \infty$ satisfies*

$$\mathrm{TV}_\mu(f) \geq c_2 \sqrt{\alpha} \, \mathrm{Var}_\mu(f) \sqrt{\log \left(1 + \frac{c_3 \alpha}{J_\mu(f)} \right)}.$$

If $J_\mu(f) = 0$, then f is constant μ -almost everywhere and the right-hand side is interpreted as zero.

The variance is forced by non-Boolean data:

$$\mathrm{Var}_\mu(f) = (\mu f)(1 - \mu f) - \int f(1 - f) \, d\mu.$$

Thus the Booleanity defect vanishes for $f = \mathbf{1}_E$, while a constant $f \equiv a \in (0, 1)$ has zero variation and zero variance. In particular, Theorem 1.2 specializes exactly to Theorem 1.1.

1.1 Proof architecture

We normalize to $\alpha = 1$ and write

$$F = \mu(E)(1 - \mu(E)), \quad P = P_\mu(E), \quad J = J_\mu(E).$$

The proof proceeds as follows.

1. A Witten-resolvent comparison proves, for every $h \in W^{1,2}(\mu)$,

$$\mathrm{Var}_\mu h \leq C \sum_i \frac{\|\partial_i h\|_2^2}{1 + \log(\|\partial_i h\|_2 / \|\partial_i h\|_1)}.$$

Only scalar hypercontractivity is used after the resolvent comparison.

2. Define the boundary second moments

$$K_i = \int_{\partial^* E} \nu_{E,i}^2 \, d\sigma_\mu, \quad \sum_i K_i = P,$$

and the crystalline seminorm $\Phi(z) = \sum_i \sqrt{K_i} |z_i|$. For every positive diagonal quadratic form D , a nonlinear heat flow associated with a mixed cotangent norm produces a function g_D having common fidelity and common Φ -variation, together with a D -weighted L^2 gradient estimate.

3. Sion's minimax theorem converts this family into one function g satisfying simultaneously

$$\begin{aligned} \|\partial_i g\|_2^2 &\leq C(P/F) K_i \quad \text{for all } i, \\ \sum_i \sqrt{K_i} \|\partial_i g\|_1 &\leq C\sqrt{PJ}. \end{aligned}$$

4. An integral representation of the Talagrand denominator and Hölder interpolation aggregate these two estimates. Applying the smooth inequality to g yields

$$\frac{P^2}{F^2} \gtrsim 1 + \log_+ \left(\frac{P}{\sqrt{FJ}} \right).$$

Gaussian isoperimetry closes the remaining logarithmic algebra.

The nonlinear flow is the only genuinely new finite-scale mechanism. It contracts the scalar anisotropic variation selected by the boundary while regularizing an independently chosen quadratic objective. It does not claim coordinatewise contraction for the Langevin semigroup.

1.2 Related analytic ingredients

The smooth influence inequality follows the hypercontractive approach of Talagrand and its continuous formulations; see [5, 6]. The covariance representation is the Helffer–Sjöstrand identity [10]. Nonlinear heat flow for Minkowski/Finsler energies and the associated weak Bochner formula were developed in [12–14]. Our use is a flat weighted specialization, but we give the calculation and verify the noncompact integrability needed here. Weighted- BV notation and approximation follow [1].

2 Preliminaries and normalization

Throughout the main proof, $\nabla^2 V \succeq I$, and $d\sigma_\mu = p_\mu \, d\mathcal{H}^{n-1}$ on $\partial^* E$. Put

$$f = \mathbf{1}_E, \quad q = \mu(E), \quad F = q(1 - q), \quad (1)$$

$$P = P_\mu(E), \quad I_i = I_i^\mu(E), \quad K_i = \int_{\partial^* E} \nu_{E,i}^2 \, d\sigma_\mu. \quad (2)$$

Since $|\nu_E|_2 = 1$ almost everywhere,

$$\sum_i K_i = P. \quad (3)$$

Also $K_i \leq I_i$, because $|\nu_{E,i}|^2 \leq |\nu_{E,i}|$.

If $0 < q < 1$, Gaussian isoperimetry under curvature one gives $P > 0$. Moreover,

$$P \leq \sum_i I_i,$$

so $J > 0$. We henceforth work in this nontrivial case.

For a norm N on cotangent vectors, its polar norm on tangent vectors is

$$N^\circ(v) = \sup_{N(z) \leq 1} v \cdot z.$$

The weighted anisotropic perimeter is

$$P_N(E) = \int_{\partial^* E} N(\nu_E) d\sigma_\mu.$$

For $u \in W_{\text{loc}}^{1,1}$, write

$$\text{TV}_N(u) = \int N(\nabla u) d\mu.$$

The same notation denotes the lower-semicontinuous weighted- BV extension.

3 A smooth Talagrand inequality without product structure

The first ingredient is an L^1 - L^2 inequality whose constants do not depend on the interaction in V .

Proposition 3.1 (Coupled smooth Talagrand inequality). *For every $h \in W^{1,2}(\mu)$, set*

$$a_i = \|\partial_i h\|_{L^1(\mu)}, \quad b_i = \|\partial_i h\|_{L^2(\mu)}.$$

Then

$$\text{Var}_\mu h \leq C \sum_{i=1}^n \frac{b_i^2}{1 + \log(b_i/a_i)}. \quad (4)$$

A zero derivative contributes zero.

Proof. Let

$$L = \Delta - \nabla V \cdot \nabla, \quad \mathcal{L} = -L,$$

and let the Witten operator on vector fields be

$$\mathcal{A} = \mathcal{L} \otimes I + \nabla^2 V.$$

Its closed form is

$$\mathfrak{a}(X, X) = \int (|\nabla X|_{\text{HS}}^2 + X^T \nabla^2 V X) d\mu.$$

If h is smooth and compactly supported, the Helffer–Sjöstrand identity gives

$$\text{Var}_\mu h = \langle \nabla h, \mathcal{A}^{-1} \nabla h \rangle_{L^2(\mu; \mathbb{R}^n)}. \quad (5)$$

For clarity, let $u = \mathcal{L}^{-1}(h - \mu h)$. The curvature-one Poincaré inequality makes the inverse well-defined on mean-zero functions. Differentiation in the weak sense gives

$$\mathcal{A} \nabla u = \nabla h,$$

and weighted integration by parts gives

$$\text{Var}_\mu h = \langle \nabla u, \nabla h \rangle,$$

which proves (5). A cutoff Reilly identity justifies the form domains when $\nabla^2 V$ is not bounded above.

Curvature gives the form order

$$\mathcal{A} \succeq (\mathcal{L} + 1) \otimes I.$$

The variational formula for a positive inverse therefore yields

$$\text{Var}_\mu h = \sum_i \langle \partial_i h, (\mathcal{L} + 1)^{-1} \partial_i h \rangle. \quad (6)$$

This is an integrated resolvent comparison, not a componentwise intertwining assertion.

Let $P_t = e^{-t\mathcal{L}}$. For scalar $r \in L^2(\mu)$,

$$\begin{aligned} \langle r, (\mathcal{L} + 1)^{-1} r \rangle &= \int_0^\infty e^{-t} \langle r, P_t r \rangle dt \\ &= \int_0^\infty e^{-t} \|P_{t/2} r\|_2^2 dt. \end{aligned} \quad (7)$$

The curvature-one logarithmic Sobolev inequality and Gross hypercontractivity [2, 9] give

$$\|P_{t/2} r\|_2 \leq \|r\|_{1+e^{-t}}.$$

Put $A = \|r\|_2$, $B = \|r\|_1$, and $H = \log(A/B) \geq 0$. Interpolation between L^1 and L^2 gives

$$\|r\|_{1+e^{-t}} \leq B^{\tanh(t/2)} A^{1-\tanh(t/2)}.$$

Consequently the right side of (7) is at most

$$A^2 \int_0^\infty e^{-t-2H \tanh(t/2)} dt.$$

With $s = \tanh(t/2)$, one has $e^{-t} dt = 2(1+s)^{-2} ds$, and hence

$$\int_0^\infty e^{-t-2H \tanh(t/2)} dt \leq 2 \int_0^1 e^{-2Hs} ds \leq \frac{C}{1+H}.$$

Substitution into (6) proves (4) for smooth compactly supported h .

For general $h \in W^{1,2}(\mu)$, multiply by radial cutoffs and then mollify on fixed balls. The cutoff error tends to zero because h and ∇h are square integrable and μ is finite. On a fixed ball, p_μ is positive and C^1 , so ordinary convolution converges in the weighted Sobolev norm. Thus $h_k \rightarrow h$ and $\partial_i h_k \rightarrow \partial_i h$ in L^2 , hence also in L^1 , and all terms pass to the limit. $\square$

4 Nonlinear Minkowski heat at finite scale

The next lemma replaces the coordinatewise regularization available in a product space. It requires only convexity of V ; the curvature-one assumption reappears in Section 3 and in the final isoperimetric closure.

Lemma 4.1 (Finite-scale Minkowski regularization). *Assume $\nabla^2 V \succeq 0$, and let N be a reversible norm on cotangent vectors that is smooth and strongly convex away from zero. Let S_t^N be the $L^2(\mu)$ -gradient flow of*

$$\mathcal{H}_N(u) = \frac{1}{2} \int N(\nabla u)^2 d\mu.$$

Let f have finite weighted N -variation and satisfy $0 \leq f \leq 1$, and put $u_t = S_t^N f$. Then, for every $t > 0$,

$$0 \leq u_t \leq 1, \quad \int u_t d\mu = \int f d\mu, \quad (8)$$

$$\mathrm{TV}_N(u_t) \leq \mathrm{TV}_N(f), \quad (9)$$

$$\|N(\nabla u_t)\|_\infty \leq (8t)^{-1/2}, \quad (10)$$

$$\|u_t - f\|_2^2 \leq \sqrt{2t} \mathrm{TV}_N(f), \quad (11)$$

$$\int N(\nabla u_t)^2 d\mu \leq \frac{\mathrm{TV}_N(f)}{\sqrt{8t}}. \quad (12)$$

If $f = \mathbf{1}_E$, then additionally

$$\|u_t - \mathbf{1}_E\|_1 \leq \sqrt{2t} P_N(E). \quad (13)$$

The displayed constants are independent of the dimension and of the ellipticity and smoothness moduli of N .

Proof, including the main identities. Set

$$\begin{aligned} H(z) &= \frac{1}{2} N(z)^2, & \mathcal{J}(z) &= DH(z), \\ B(z) &= D^2 H(z) & (z \neq 0). \end{aligned}$$

The functional $\mathcal{H}_N$ is proper, closed, convex, and Markovian on $L^2(\mu)$. Maximal-monotone gradient-flow theory [4, 13] gives a unique contraction flow satisfying

$$\partial_t u = \mathrm{div}_\mu \mathcal{J}(\nabla u) \quad (14)$$

weakly. Scalar truncation proves the range bound. Translation invariance of $\mathcal{H}_N$ makes every subgradient orthogonal to constants, and hence preserves the mean. This proves (8).

At a regular point with $\nabla u \neq 0$, define the linearization

$$L_u h = \mathrm{div}_\mu (B(\nabla u) \nabla h).$$

Euler homogeneity gives

$$B(z)z = \mathcal{J}(z), \quad z \cdot \mathcal{J}(z) = N(z)^2. \quad (15)$$

Writing $U = D^2 u$, direct differentiation of (14) and the diffusion chain rule give

$$(\partial_t - L_u)H(\nabla u) = -Q - \mathcal{J}(\nabla u)^T \nabla^2 V \mathcal{J}(\nabla u) \leq 0, \quad (16)$$

where

$$Q = \mathrm{tr} \left[(B^{1/2} U B^{1/2})^2 \right] \geq 0.$$

The exact constant-one Kato estimate follows from

$$\begin{aligned} \Gamma_B(H) &= |B^{1/2} U \mathcal{J}|^2 \\ &\leq Q |B^{-1/2} \mathcal{J}|^2 = Q N(\nabla u)^2. \end{aligned} \quad (17)$$

The chain rule for $\sqrt{2H + \delta^2}$, followed by $\delta \downarrow 0$, therefore gives

$$(\partial_t - L_u)N(\nabla u) \leq 0, \quad (\partial_t - L_u)N(\nabla u)^2 \leq 0. \quad (18)$$

There are two analytic points hidden by a formal smooth calculation: H is not C^2 at zero unless N is Hilbertian, and the domain is noncompact. In Appendix A we prove the distributional extension through $\{\nabla u = 0\}$, verify the required global intrinsic integrability by a cut-off Reilly identity, and construct the conservative whole-space linearized evolution. In particular there is a forward Markov propagator $U(t, s)$ for $\partial_t - L_{u_t}$ such that comparison in (18) is valid and

$$\int U(t, s) h d\mu = \int h d\mu.$$

Integrating the first subsolution inequality proves (9) for smooth Sobolev data.

For the pointwise estimate, (15) gives

$$(\partial_t - L_u)(u - c)^2 = -2N(\nabla u)^2. \quad (19)$$

Duhamel's formula, with the forward orientation, reads

$$U(t, 0)(u_0 - c)^2 - (u_t - c)^2 = 2 \int_0^t U(t, s) N(\nabla u_s)^2 ds. \quad (20)$$

The second inequality in (18) implies

$$N(\nabla u_t)^2 \leq U(t, s) N(\nabla u_s)^2.$$

If $0 \leq u_0 \leq 1$, take $c = 1/2$. Positivity and the Markov property in (20) give

$$2t N(\nabla u_t)^2 \leq \frac{1}{4},$$

which proves (10).

We now pass to weighted BV data. As proved in Appendix B, there are $f_k \in C_c^\infty(\mathbb{R}^n)$, $0 \leq f_k \leq 1$, such that

$$f_k \rightarrow f \text{ in } L^1(\mu) \cap L^2(\mu), \quad \int N(\nabla f_k) d\mu \rightarrow \mathrm{TV}_N(f). \quad (21)$$

Let $u_t^k = S_t^N f_k$. Nonlinear L^2 -contraction, lower semicontinuity, and the weak-star closedness of the N -gradient ball pass (9) and (10) to u_t .

The noise estimate needs neither nonlinear self-adjointness nor a boundary trace. Since

$$N^\circ(\mathcal{J}(z)) = N(z), \quad (22)$$

testing the weak equation for u_s^k against the fixed function f_k gives

$$\begin{aligned} |\langle f_k - u_t^k, f_k \rangle| &\leq \int_0^t \|N(\nabla u_s^k)\|_\infty \, ds \int N(\nabla f_k) \, d\mu \\ &\leq \sqrt{t/2} \int N(\nabla f_k) \, d\mu. \end{aligned}$$

Pass to the limit using (21). Since $S_t^N 0 = 0$, L^2 -contraction gives $\|u_t\|_2 \leq \|f\|_2$. Consequently,

$$\|u_t - f\|_2^2 \leq 2 \langle f - u_t, f \rangle \leq \sqrt{2t} \, \text{TV}_N(f),$$

which proves (11). When $f = \mathbf{1}_E$, range and mean preservation also imply the exact identity

$$\|u_t - \mathbf{1}_E\|_1 = 2 \langle \mathbf{1}_E - u_t, \mathbf{1}_E \rangle,$$

which proves (13). Finally,

$$\int N(\nabla u_t)^2 \, d\mu \leq \|N(\nabla u_t)\|_\infty \, \text{TV}_N(u_t),$$

and (9)–(10) give (12). $\square$

Remark 4.2 (Why this does not contradict the mixed-norm obstruction). The flow in Lemma 4.1 changes with the chosen norm. It contracts one scalar anisotropic variation; it does not assert $J(S_t \mathbf{1}_E) \lesssim J(E)$ for the Langevin semigroup. The latter would confuse $\ell_2(L^1)$ with $L^1(\ell_2)$ and is false in coupled Gaussian examples.

5 A set-dependent bridge for every quadratic objective

Define the crystalline seminorm

$$\Phi(z) = \sum_{i=1}^n \sqrt{K_i} |z_i|. \quad (23)$$

Its boundary variation is

$$P_\Phi(E) = \sum_i \sqrt{K_i} I_i \leq \sqrt{\sum_i K_i} \sqrt{\sum_i I_i^2} = \sqrt{PJ}. \quad (24)$$

For a nontrivial set this is positive: some $K_i > 0$, and then $I_i \geq K_i > 0$.

Fix a positive diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$, and put

$$P_D = \int_{\partial^* E} \sqrt{\nu_E^T D \nu_E} \, d\sigma_\mu, \quad (25)$$

$$K_D = \sum_i d_i K_i. \quad (26)$$

Boundary Cauchy–Schwarz gives

$$P_D^2 \leq PK_D. \quad (27)$$

Set

$$\eta_D = \frac{P_D}{P_\Phi(E)}, \quad \Psi_D(z) = \sqrt{z^T D z + \eta_D^2 \Phi(z)^2}. \quad (28)$$

Then

$$P_{\Psi_D}(E) \leq P_D + \eta_D P_\Phi(E) = 2P_D. \quad (29)$$

The norm Ψ_D need not be smooth. By standard convex-body smoothing, choose a reversible smooth strongly convex norm N_D satisfying

$$\frac{1}{2} \Psi_D \leq N_D \leq 2\Psi_D. \quad (30)$$

For completeness, add a sufficiently small Euclidean ball to the symmetric unit ball, smooth and symmetrize its support function on the sphere, and pass to the polar body. Taking the approximation sufficiently close gives (30) in every finite dimension. The ellipticity of the resulting norm may depend on D , but no estimate below uses it. Consequently,

$$\begin{aligned} N_D(z) &\geq \frac{1}{2} \sqrt{z^T D z}, \quad N_D(z) \geq \frac{1}{2} \eta_D \Phi(z), \\ P_{N_D}(E) &\leq 4P_D. \end{aligned} \quad (31)$$

Choose a small universal $\theta > 0$, let

$$\sqrt{t_D} = \theta \frac{F}{P_{N_D}(E)}, \quad g_D = S_{t_D}^{N_D} f. \quad (32)$$

By Lemma 4.1, after fixing θ ,

$$0 \leq g_D \leq 1, \quad \mu g_D = q, \quad \|g_D - f\|_1 \leq F/16. \quad (33)$$

The total-variation estimate and (31) give the common crystalline constraint

$$\sum_i \sqrt{K_i} \|\partial_i g_D\|_1 = \text{TV}_\Phi(g_D) \leq CP_\Phi(E) \leq C\sqrt{PJ}. \quad (34)$$

Likewise, using Equations (12) and (27),

$$\begin{aligned} \sum_i d_i \|\partial_i g_D\|_2^2 &\leq C \int N_D(\nabla g_D)^2 \, d\mu \\ &\leq C \frac{P_{N_D}(E)^2}{F} \\ &\leq C \frac{P}{F} \sum_i d_i K_i. \end{aligned} \quad (35)$$

Thus every positive diagonal objective has a comparison function satisfying the same range, mean, fidelity, and crystalline constraints.

6 Minimax synchronization

Put $\lambda = P/F$. For $0 < \varepsilon < 1/2$, define

$$\Delta_\varepsilon = \left\{ d_i \geq \frac{\varepsilon}{n} : \sum_i d_i = 1 \right\}.$$

If $d \in \Delta_\varepsilon$, then Equation (35) and $K_D \leq P$ imply

$$\sum_i \|\partial_i g_D\|_2^2 \leq C\lambda \frac{nP}{\varepsilon}. \quad (36)$$

Let X_ε be the set of $g \in W^{1,2}(\mu)$ satisfying Equations (33), (34) and (36), with harmless fixed enlargements of the constants. It is convex and weakly compact. Indeed, the range controls the L^2 norm; the mean constraint is affine; and fidelity, crystalline variation, and quadratic energy are convex weakly lower-semicontinuous functionals. Every g_D belongs to this same set.

Define

$$\mathcal{Q}(g, d) = \sum_i d_i \left(\|\partial_i g\|_2^2 - C\lambda K_i \right). \quad (37)$$

The function is convex and weakly lower semicontinuous in g , and affine continuous in d . Sion's theorem [15] and Equation (35) give

$$\min_{g \in X_\varepsilon} \max_{d \in \Delta_\varepsilon} \mathcal{Q}(g, d) = \max_{d \in \Delta_\varepsilon} \min_{g \in X_\varepsilon} \mathcal{Q}(g, d) \leq 0.$$

Choose a minimizer g_ε , and set

$$v_i = \|\partial_i g_\varepsilon\|_2^2 - C\lambda K_i.$$

Since

$$\max_{d \in \Delta_\varepsilon} d \cdot v = (1 - \varepsilon) \max_i v_i + \frac{\varepsilon}{n} \sum_i v_i$$

and $\sum_i v_i \geq -C\lambda P$, we get

$$\|\partial_i g_\varepsilon\|_2^2 \leq C\lambda K_i + \frac{C\varepsilon}{1 - \varepsilon} \frac{\lambda P}{n}. \quad (38)$$

Summing gives an ε -independent Sobolev bound. Along a sequence $\varepsilon \downarrow 0$, weak compactness and lower semicontinuity give a single $g \in W^{1,2}(\mu)$ such that

$$0 \leq g \leq 1, \quad \mu g = q, \quad \|g - f\|_1 \leq F/16, \quad (39)$$

$$\|\partial_i g\|_2^2 \leq C \frac{P}{F} K_i \quad (1 \leq i \leq n), \quad (40)$$

$$\sum_i \sqrt{K_i} \|\partial_i g\|_1 \leq C\sqrt{PJ}. \quad (41)$$

Because g and f have the same mean and $|g - f| \leq 1$,

$$\sqrt{\text{Var}_\mu g} \geq \sqrt{F} - \|g - f\|_2 \geq \frac{3}{4}\sqrt{F}. \quad (42)$$

7 Moment aggregation and proof of the main theorem

Set

$$a_i = \|\partial_i g\|_1, \quad b_i = \|\partial_i g\|_2, \quad A_0 = \sum_i \sqrt{K_i} a_i.$$

For a nonzero term,

$$\frac{1}{1 + \log(b_i/a_i)} = \int_0^\infty e^{-s} (a_i/b_i)^s ds. \quad (43)$$

For $0 \leq s \leq 1$, Equation (40) and Hölder give

$$\begin{aligned} \sum_i b_i^{2-s} a_i^s &\leq C\lambda^{1-s/2} \sum_i K_i^{1-s} (\sqrt{K_i} a_i)^s \\ &\leq C\lambda^{1-s/2} P^{1-s} A_0^s. \end{aligned} \quad (44)$$

By Equation (41), $A_0 \leq C\sqrt{PJ}$. Therefore, with

$$R = \frac{P}{\sqrt{FJ}},$$

the last expression is at most

$$C \frac{P^2}{F} R^{-s}. \quad (45)$$

For $s \geq 1$, the moment is at most its value at one because $a_i \leq b_i$. Integrating Equations (43) and (45), and using the $s = 0$ estimate when $R < 1$, yields

$$\sum_i \frac{b_i^2}{1 + \log(b_i/a_i)} \leq \frac{C(P^2/F)}{1 + \log_+(P/\sqrt{FJ})}. \quad (46)$$

Apply Proposition 3.1 to g and use Equations (42) and (46). If

$$X = \frac{P^2}{F^2},$$

then

$$X \geq c \left[1 + \log_+ \left(\frac{P}{\sqrt{FJ}} \right) \right] = c \left[1 + \frac{1}{2} \log_+ \left(\frac{XF}{J} \right) \right]. \quad (47)$$

The Bakry–Ledoux Gaussian isoperimetric comparison under curvature one [3] gives both the Cheeger bound $X \geq c_*$ and

$$X \geq c[1 + \log(1/F)]. \quad (48)$$

Using $X \geq c_*$ inside (47) gives

$$X \geq c[1 + \log_+(F/J)]. \quad (49)$$

If $J \leq F$, the logarithms in Equations (48) and (49) add, up to universal constants, to $\log(1/J)$. If $J > F$, Equation (48) alone controls $\log(1 + 1/J)$. Thus

$$\frac{P^2}{F^2} \geq c \left[1 + \log \left(1 + \frac{c}{J} \right) \right]. \quad (50)$$

Taking square roots proves Theorem 1.1 when $\alpha = 1$.

For general α , apply the curvature-one statement after $y = \sqrt{\alpha}x$. The transformed quantities satisfy

$$\begin{aligned} \bar{P} &= \alpha^{-1/2} P, & \bar{I}_i &= \alpha^{-1/2} I_i, \\ \bar{J} &= \alpha^{-1} J, & \bar{F} &= F. \end{aligned}$$

Multiplying the resulting estimate by $\sqrt{\alpha}$ proves the theorem. The cases $F = 0$ are trivial, completing the proof.

8 The functional Eldan–Gross inequality

We now prove Theorem 1.2. This is not a formal substitution in the set theorem: coarea places the square defining J_μ on the wrong side of the level integral. The nonlinear-flow bridge supplies the missing simultaneous control.

8.1 The exact level-set consequence

Let $f \in BV_{\text{loc}}(\mathbb{R}^n)$, $0 \leq f \leq 1$, and put

$$\begin{aligned} E_t &= \{f > t\}, & q_t &= \mu(E_t), \\ F_t &= q_t(1 - q_t), & J_t &= J_\mu(E_t). \end{aligned}$$

Weighted scalar and coordinatewise coarea give

$$\text{TV}_\mu(f) = \int_0^1 P_\mu(E_t) dt, \quad I_i^\mu(f) = \int_0^1 I_i^\mu(E_t) dt. \quad (51)$$

Consequently Theorem 1.1 yields the exact level-set inequality

$$\text{TV}_\mu(f) \geq c_0 \sqrt{\alpha} \int_0^1 F_t \sqrt{\log \left(1 + \frac{c_1 \alpha}{J_t} \right)} dt. \quad (52)$$

Levels with $F_t = 0$ contribute zero. The integrated balance has the probabilistic identity

$$\int_0^1 F_t dt = \frac{1}{2} \iint |f(x) - f(y)| d\mu(x) d\mu(y).$$

However, if $a(t) = (I_i^\mu(E_t))_{i=1}^n$, then

$$\sqrt{J_\mu(f)} = \left| \int_0^1 a(t) dt \right|_2 \leq \int_0^1 \sqrt{J_t} dt.$$

This is the unfavorable direction for replacing J_t in (52) by $J_\mu(f)$. Different levels may place their boundary variation in different coordinates.

8.2 A functional crystalline bridge

It suffices first to take $\alpha = 1$. Write

$$\begin{aligned} v &= \text{Var}_\mu(f), & P &= \text{TV}_\mu(f), \\ I_i &= I_i^\mu(f), & J &= J_\mu(f). \end{aligned}$$

If $v = 0$, then f is constant and the assertion is trivial. Otherwise polar-decompose $Df = \sigma |Df|$, and write $d\omega = p_\mu d|Df|$. Thus $|\sigma|_2 = 1$ ω -almost everywhere. Define

$$K_i = \int \sigma_i^2 d\omega.$$

Then

$$\sum_i K_i = P, \quad K_i \leq I_i = \int |\sigma_i| d\omega. \quad (53)$$

For the crystalline seminorm $\Phi(z) = \sum_i \sqrt{K_i} |z_i|$,

$$\text{TV}_\Phi(f) = \sum_i \sqrt{K_i} I_i \leq \sqrt{PJ}. \quad (54)$$

For a positive diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$, put

$$P_D(f) = \int \sqrt{\sigma^T D \sigma} d\omega, \quad K_D = \sum_i d_i K_i.$$

Measure Cauchy–Schwarz gives

$$P_D(f)^2 \leq PK_D. \quad (55)$$

Since $v > 0$, one has $P > 0$; hence some $K_i > 0$, and then $I_i \geq K_i$. Thus $\text{TV}_\Phi(f) > 0$, and all denominators below are nonzero.

Repeat the construction of Section 5: mix the D -ellipsoid with Φ , smooth the resulting unit ball, and obtain N_D satisfying the analogues of Equation (31),

$$\begin{aligned} N_D(z) &\geq \frac{1}{2} \sqrt{z^T D z}, & N_D(z) &\geq \frac{1}{2} \eta_D \Phi(z), \\ \text{TV}_{N_D}(f) &\leq 4P_D(f). \end{aligned}$$

where $\eta_D = P_D(f) / \text{TV}_\Phi(f)$. Choose

$$\sqrt{t_D} = \theta \frac{v}{\text{TV}_{N_D}(f)}, \quad g_D = S_{t_D}^{N_D} f, \quad (56)$$

with $\theta > 0$ universal and sufficiently small. The general noise estimate (11) gives

$$\|g_D - f\|_2^2 \leq \sqrt{2} \theta v.$$

The flow preserves the mean, so

$$\sqrt{\text{Var}_\mu(g_D)} \geq \sqrt{v} - \|g_D - f\|_2 \geq \frac{3}{4} \sqrt{v}. \quad (57)$$

Variation contraction and Equation (54) give the common constraint

$$\sum_i \sqrt{K_i} \|\partial_i g_D\|_1 \leq C \sqrt{PJ}. \quad (58)$$

Using Equations (12) and (55) gives, independently of D ,

$$\sum_i d_i \|\partial_i g_D\|_2^2 \leq C \frac{P}{v} \sum_i d_i K_i. \quad (59)$$

Apply the minimax argument of Section 6 with the common range and mean constraints $0 \leq g \leq 1$, $\mu g = \mu f$, and with the convex fidelity constraint $\|g - f\|_2^2 \leq v/16$ in place of the indicator L^1 -constraint. It produces one $g \in W^{1,2}(\mu)$ satisfying

$$\text{Var}_\mu(g) \geq cv, \quad (60)$$

$$\|\partial_i g\|_2^2 \leq C(P/v) K_i, \quad (61)$$

$$\sum_i \sqrt{K_i} \|\partial_i g\|_1 \leq C \sqrt{PJ}. \quad (62)$$

The moment calculation of Section 7, now with $F = v$, and Proposition 3.1 imply, for $X = P^2/v^2$,

$$X \geq c \left[1 + \frac{1}{2} \log_+ \left(\frac{Xv}{J} \right) \right]. \quad (63)$$

8.3 Functional small-variance closure

We record the bounded-function version of Gaussian small-mass isoperimetry.

Lemma 8.1 (Functional Gaussian BV bound). *If $\nabla^2 V \succeq I$, $0 \leq f \leq 1$, and $v = \text{Var}_\mu(f)$, then*

$$\text{TV}_\mu(f) \geq cv\sqrt{1 + \log(1/v)}. \quad (64)$$

Proof. The case $v = 0$ is immediate. Assume $v > 0$. Choose a median m of f and set $h_+ = (f - m)_+$, $h_- = (m - f)_+$. Each is supported on a set of measure at most $1/2$, and the BV truncation-coarea identity gives $\text{TV}_\mu(f) = \text{TV}_\mu(h_+) + \text{TV}_\mu(h_-)$.

For either function h , put $r(s) = \mu(h > s)$. Gaussian isoperimetric comparison and coarea give

$$\text{TV}_\mu(h) \geq c \int_0^1 r(s) \sqrt{\log(e/r(s))} ds. \quad (65)$$

We use the elementary tail estimate

$$\int_0^1 r(s) \sqrt{\log(e/r(s))} ds \geq c \mu(h^2) \sqrt{\log(e/\mu(h^2))}. \quad (66)$$

If $h = 0$, this is trivial. Otherwise, let $p_k = \mu(2^{-k-1} < h \leq 2^{-k})$ and $x_k = 4^{-k} p_k$. On $(2^{-k-2}, 2^{-k-1})$, one has $r(s) \geq p_k$; hence the left side of (66) is at least

$$c \sum_{k \geq 0} 2^{-k} p_k \sqrt{\log(e/p_k)} \geq c \sum_{k \geq 0} x_k \sqrt{\log(e/x_k)}.$$

For the second inequality, set $a = 2^{-k}$ and $L = \log(e/p_k) \geq 1$. Since

$$a^{-2}L \geq L + 2\log(1/a) = \log\left(\frac{e}{p_k a^2}\right),$$

each term on the first sum dominates the corresponding term on the second. Zero p_k 's are omitted, or interpreted by continuity. Since $\mu(h^2) \leq \sum_k x_k \leq 4\mu(h^2)$, and $\sqrt{\log(e/x_k)} \geq \sqrt{\log(e/\sum_j x_j)}$, the second sum is at least

$$S \sqrt{\log(e/S)}, \quad S = \sum_k x_k.$$

Moreover, $S \leq \sum_k p_k \leq 1/2$. The map $x \mapsto x \sqrt{\log(e/x)}$ is increasing on $[0, 1]$; since $S \geq \mu(h^2)$, this proves (66).

Finally,

$$\mu(h_+^2) + \mu(h_-^2) = \mu((f - m)^2) \geq \text{Var}_\mu(f).$$

The map $\phi(x) = x \sqrt{\log(e/x)}$ is increasing, and $\phi(x)/x$ is decreasing, so ϕ is superadditive on $[0, 1]$. Thus

$$\phi(\mu(h_+^2)) + \phi(\mu(h_-^2)) \geq \phi(\mu((f - m)^2)) \geq \phi(v),$$

which proves (64). $\square$

The lemma gives

$$X \geq c[1 + \log(1/v)].$$

Using the universal lower bound for X inside (63) also gives

$$X \geq c[1 + \log_+(v/J)].$$

If $J \leq v$, these two logarithms add to $\log(1/J)$; if $J > v$, the first already controls $\log(1 + 1/J)$. Therefore

$$\frac{P^2}{v^2} \geq c \left[1 + \log\left(1 + \frac{c}{J}\right) \right].$$

This proves Theorem 1.2 for $\alpha = 1$. Under $y = \sqrt{\alpha}x$, P and every I_i scale by $\alpha^{-1/2}$, J scales by α^{-1} , and v is unchanged. The stated inequality follows.

Corollary 8.2 (Oscillation-normalized form). *Under the measure hypotheses of Theorem 1.2, let f be bounded with finite weighted BV variation and essential oscillation $r > 0$. Then*

$$\text{TV}_\mu(f) \geq c\sqrt{\alpha} \frac{\text{Var}_\mu(f)}{r} \sqrt{\log\left(1 + \frac{c\alpha r^2}{J_\mu(f)}\right)}.$$

Proof. Apply Theorem 1.2 to $(f - \text{ess inf } f)/r$. $\square$

9 Discussion

The construction isolates two principles that may be useful beyond the present inequality. First, curvature-only smooth influence inequalities can be obtained from one-form resolvent domination even when coordinate derivatives do not commute with the diffusion. Second, when different quadratic objectives require different regularizers, a common scalar variation constraint can make the family compatible with minimax. Here the boundary second moments K_i determine the crystalline common currency.

The fixed coordinate system is essential to the statement and is never rotated. The proof also introduces no condition number: the auxiliary norm may be extremely anisotropic, but its ellipticity constants are used only to construct its flow and linearized propagator. Every displayed estimate is independent of those constants.

A Weak whole-space Minkowski heat

This appendix supplies the analytic details behind Lemma 4.1. Throughout, N is a fixed smooth strongly convex reversible cotangent norm, $H = N^2/2$, $\mathcal{J} = DH$, and $\nabla^2 V \succeq 0$. Constants used only to construct weak solutions may depend on the equivalence between N and the Euclidean norm; none survives in the estimates of Lemma 4.1.

A.1 Abstract flow and positive-time regularity

Define

$$W^{1,2}(\mu) = \{u \in L^2(\mu) : \nabla u \in L^2(\mu; \mathbb{R}^n)\},$$

with distributional derivatives. Smooth compactly supported functions plus constants are dense. Indeed, radial cutoffs χ_R with $|\nabla \chi_R| \leq C/R$ converge to one in the form norm because μ is finite, and ordinary mollification works locally since the density is positive and C^1 .

The functional

$$\mathcal{H}_N(u) = \frac{1}{2} \int N(\nabla u)^2 d\mu, \quad u \in W^{1,2}(\mu),$$

extended by $+\infty$ outside $W^{1,2}$, is lower semicontinuous and convex. If $T : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar normal contraction, the Sobolev chain rule gives

$$N(\nabla(T \circ u)) \leq N(\nabla u) \quad \text{a.e.}$$

Tests of the resolvent equation by the negative part and by $(u-1)_+$ show that the flow is order and range preserving. Applying the subgradient inequality to $u+c$ with both signs of c gives

$$\int \xi d\mu = 0 \quad (\xi \in \partial \mathcal{H}_N(u)),$$

which proves mean preservation. Standard maximal-monotone theory also gives

$$\|S_t^N f - S_t^N g\|_2 \leq \|f - g\|_2, \quad (67)$$

$$\partial_t u_t \in -\partial \mathcal{H}_N(u_t) \quad \text{for a.e. } t > 0, \quad (68)$$

$$\frac{d}{dt} \mathcal{H}_N(u_t) = -\|\partial_t u_t\|_2^2 \quad \text{for a.e. } t > 0. \quad (69)$$

For smooth compactly supported initial data, the weak equation is

$$\int \partial_t u \varphi d\mu = - \int \mathcal{J}(\nabla u) \cdot \nabla \varphi d\mu. \quad (70)$$

Since $D\mathcal{J}$ is bounded and uniformly positive definite away from zero, with a bounded elliptic measurable extension at zero, spatial difference quotients and time Steklov averages give at positive regular times

$$u \in H_{\text{loc}}^2(\mathbb{R}^n), \quad \partial_t u \in H_{\text{loc}}^1(\mathbb{R}^n).$$

One may equivalently use the local $C^{1,\beta}$ regularity of nonlinear Minkowski heat from [13]. We do not assert that u is C^2 .

A.2 The zero-gradient set and weak Bochner–Kato

For $z \neq 0$, put $B(z) = D^2 H(z)$. Its eigenvalues are bounded above and below on the Euclidean unit sphere.

Choose any measurable positive definite value $B(0)$ between these bounds. Each component $v_i = \partial_i u$ lies locally in H^1 . The Stampacchia level-set lemma gives $\nabla v_i = 0$ almost everywhere on $\{v_i = 0\}$, and hence

$$D^2 u = 0 \quad \text{a.e. on } \{\nabla u = 0\}. \quad (71)$$

Thus $B(0)D^2 u$, and all weak chain rules below, are independent of the chosen extension.

On $\{\nabla u \neq 0\}$, differentiation of the equation gives

$$\partial_t(\partial_k u) = L_u(\partial_k u) - (\nabla^2 V \mathcal{J}(\nabla u))_k. \quad (72)$$

The divergence-form chain rule applied to $H(\nabla u)$ yields

$$(\partial_t - L_u)H(\nabla u) = -Q - \mathcal{J}^T \nabla^2 V \mathcal{J}, \quad (73)$$

where

$$Q = \text{tr}[(B^{1/2} D^2 u B^{1/2})^2].$$

No third derivative of H remains: the derivatives of B occurring in the differentiated flux and in the chain rule cancel.

Let $r = N(\nabla u)$. If $C = B^{1/2} D^2 u B^{1/2}$ and $z = B^{-1/2} \mathcal{J} = B^{1/2} \nabla u$, then $|z| = r$ and

$$\Gamma_B(H) = |Cz|^2 \leq r^2 \text{tr}(C^2) = r^2 Q. \quad (74)$$

For $r_\delta = \sqrt{r^2 + \delta^2} = \sqrt{2H + \delta^2}$, the weak chain rule therefore gives, away from $r = 0$,

$$(\partial_t - L_u)r_\delta \leq -\frac{Q}{r_\delta} + \frac{Qr^2}{r_\delta^3} - \frac{\mathcal{J}^T \nabla^2 V \mathcal{J}}{r_\delta} \leq 0.$$

We explain the passage through $r = 0$. The Sobolev level-set lemma also gives $\nabla r = 0$ almost everywhere on $\{r = 0\}$. For a nonnegative compactly supported bounded Sobolev test function φ , use

$$\varphi_k = \min\{\varphi, k^2 r^2\}.$$

It vanishes on the critical set and is admissible in the identity on $\{r > 0\}$. On $\Omega_k = \{r > 0 : \varphi > k^2 r^2\}$, one has $k^2 r^2 \leq \varphi$. The only potentially bad cutoff term is bounded by

$$2 \int_{\Omega_k} \varphi \Gamma_B(r) d\mu, \quad (75)$$

which tends to zero once the global integrability proved in the next subsection is available; Ω_k decreases to a null subset of $\{r = 0\}$. The remaining terms converge by monotone convergence and (71). We obtain (73) distributionally on all of space. Applying the same argument to r_δ , then sending $\delta \downarrow 0$, proves

$$(\partial_t - L_u)r \leq 0, \quad (\partial_t - L_u)r^2 \leq 0 \quad (76)$$

in the weak, respectively truncated- L^1 , senses.

Geometrically, the tangent norm corresponding to the cotangent Hamiltonian H is N° . Its geodesics are straight lines and its fundamental tensor is independent of the base point. Thus the weighted Finsler curvature term is exactly $\nabla^2 V[v, v] \geq 0$. The auxiliary flow uses only nonnegative curvature; the Euclidean condition $\nabla^2 V \succeq I$ does not imply curvature one relative to an arbitrary auxiliary norm.

A.3 The cutoff Reilly identity

The noncompact comparison argument requires $\Gamma_B(r) \in L^1(\mu)$. This is a consequence of the equation. Put

$$w = \partial_t u = \operatorname{div}_\mu \mathcal{J}(\nabla u).$$

At a positive regular time, $w \in L^2(\mu)$. Let χ_R be radial, equal to one on B_R , zero outside B_{2R} , and satisfy $|\nabla \chi_R| \leq C/R$. Multiply (72) by $\chi_R^2 \mathcal{J}_k(\nabla u)$, sum over k , and integrate twice by parts. This gives

$$\int \chi_R^2 w^2 d\mu = \int \chi_R^2 (Q + \mathcal{J}^T \nabla^2 V \mathcal{J}) d\mu + \operatorname{Err}_R. \quad (77)$$

The two errors have the forms

$$\int \chi_R w \mathcal{J} \cdot \nabla \chi_R d\mu, \quad \int \chi_R \nabla \chi_R^T B(D^2 u) \mathcal{J} d\mu.$$

Polar duality and Equation (74) give

$$|\mathcal{J} \cdot \nabla \chi_R| \leq r N(\nabla \chi_R), \\ |\nabla \chi_R^T B(D^2 u) \mathcal{J}| \leq C_N |\nabla \chi_R| r \sqrt{Q}.$$

Young's inequality absorbs half of the localized Q term and gives

$$\int \chi_R^2 (Q + \mathcal{J}^T \nabla^2 V \mathcal{J}) d\mu \leq C_N \|w\|_2^2 + \frac{C_N}{R^2} \|r\|_2^2.$$

Thus the two nonnegative terms are globally integrable. Returning to (77), Cauchy–Schwarz and tail exhaustion make both errors tend to zero, yielding

$$\int (Q + \mathcal{J}^T \nabla^2 V \mathcal{J}) d\mu = \|\partial_t u\|_2^2. \quad (78)$$

Since $\Gamma_B(r) \leq Q$, this proves the required integrability and justifies the limit in (75). Also $r^2 \in L^1$ and $\sqrt{\Gamma_B(r^2)} \leq 2r\sqrt{Q} \in L^1$, which puts r^2 in the truncated comparison class; we do not claim $r^2 \in W^{1,2}$.

A.4 The conservative forward propagator

At positive times consider the measurable symmetric forms

$$\mathfrak{b}_s(h, k) = \int \nabla h^T B(\nabla u_s) \nabla k d\mu$$

with common domain $W^{1,2}(\mu)$. The fixed norm supplies finite upper and lower ellipticity constants, so the theory of measurable nonautonomous coercive forms constructs an evolution family $U(t, s)$, $0 \leq s \leq t$, satisfying

$$\partial_t U(t, s)h = L_{u_t} U(t, s)h, \quad U(s, s) = I.$$

Positive-part and unit-truncation tests show that it is positivity preserving and L^∞ -contractive. Since $1 \in W^{1,2}(\mu)$ and $\mathfrak{b}_s(1, h) = 0$,

$$U(t, s)1 = 1, \quad \int U(t, s)h d\mu = \int h d\mu. \quad (79)$$

No boundary at infinity occurs in this construction.

If z is a weak subsolution of $\partial_t z - L_{u_t} z \leq 0$, Steklov averaging and testing the positive part of $z_t - U(t, s)z_s$ give

$$z_t \leq U(t, s)z_s. \quad (80)$$

Truncation and monotone convergence extend this to the L^1 subsolution r^2 . Applying (80) to r , then using (79), gives total-variation contraction.

For an energy-class path h_s , differentiation in the lower endpoint is

$$\partial_s [U(t, s)h_s] = U(t, s)(\partial_s h_s - L_{u_s} h_s).$$

Because $B(\nabla u) \nabla u = \mathcal{J}(\nabla u)$, the nonlinear solution satisfies $\partial_t u = L_{u_t} u$. Apply the preceding identity to $h_s = u_s^2$, for which

$$(\partial_s - L_{u_s})u_s^2 = -2r_s^2.$$

This yields

$$U(t, 0)f^2 - u_t^2 = 2 \int_0^t U(t, s)r_s^2 ds. \quad (81)$$

The propagator is forward; it is not asserted to be self-adjoint as a two-parameter family. Since $r_t^2 \leq U(t, s)r_s^2$, $U(t, 0)f = u_t$, and $f^2 \leq f$ for $0 \leq f \leq 1$,

$$2tr_t^2 \leq u_t - u_t^2 \leq \frac{1}{4}.$$

All identities are first obtained for time-regularized solutions and almost every positive time. Semigroup continuity, lower semicontinuity, and the evolution law extend the estimates to every $t > 0$.

B Strict weighted-BV approximation

We prove (21) for every $h \in BV_{\text{loc}}(\mathbb{R}^n)$ with $0 \leq h \leq 1$ and finite weighted variation. The weighted anisotropic variation is

$$|Dh|_{N, \mu} = \int p_\mu(x) N \left(\frac{dDh}{d|Dh|} \right) d|Dh|.$$

For $h = \mathbf{1}_E$, this equals $P_N(E)$.

Let χ_R be the radial cutoff used above. The BV product rule gives

$$|D(\chi_R h)|_{N, \mu} \leq \int \chi_R p_\mu N \left(\frac{dDh}{d|Dh|} \right) d|Dh| \\ + \int |h| N(\nabla \chi_R) d\mu.$$

The first term tends to $|Dh|_{N, \mu}$, while the second is bounded by

$$\frac{C_N}{R} \mu(B_{2R} \setminus B_R) \longrightarrow 0.$$

Since $\chi_R h \rightarrow h$ in $L^1(\mu)$, lower semicontinuity shows that the variations converge as well.

Fix R . On B_{2R} , the density p_μ is positive and uniformly continuous. Ordinary strict Euclidean BV approximation followed by positive convolution produces $h_{R,k} \in C_c^\infty$, $0 \leq h_{R,k} \leq 1$, converging in L^1 . Reshetnyak continuity applied to the continuous one-homogeneous integrand

$$(x, z) \longmapsto p_\mu(x)N(z)$$

gives convergence of the weighted anisotropic variations. A diagonal choice in R, k produces $f_k \in C_c^\infty(\mathbb{R}^n)$, $0 \leq f_k \leq 1$, satisfying (21). Because all functions lie in $[0, 1]$,

$$\|f_k - h\|_2^2 \leq \|f_k - h\|_1,$$

so the convergence is also in L^2 .

The same construction justifies every weighted-BV passage in the main proof. In particular, testing the weak nonlinear equation against the smooth approximants and passing by L^2 -contraction avoids any appeal to a classical normal trace on $\partial^* E$.

C Two elementary closure lemmas

We isolate the algebraic steps used in Sections 6 and 7.

Lemma C.1 (Simplex synchronization). *Let $K_i \geq 0$, $\sum_i K_i = P$, and suppose that for every positive probability vector d there is an element g_d of a common convex set X satisfying*

$$\sum_i d_i B_i(g_d) \leq C\lambda \sum_i d_i K_i,$$

where each B_i is nonnegative, convex, and lower semicontinuous. If the truncated common sets are weakly compact, then there is $g \in X$ such that

$$B_i(g) \leq C\lambda K_i \quad \text{for every } i.$$

Proof. Apply Sion's theorem on $\Delta_\varepsilon = \{d_i \geq \varepsilon/n, \sum_i d_i = 1\}$ to $\sum_i d_i (B_i(g) - C\lambda K_i)$. The exact maximum over Δ_ε is

$$(1 - \varepsilon) \max_i v_i + \frac{\varepsilon}{n} \sum_i v_i.$$

The nonnegativity of the B_i 's bounds $\sum_i v_i$ below, giving an $O(\varepsilon/n)$ error in every coordinate. Sum the estimates to obtain compactness independent of ε , and pass weakly as $\varepsilon \downarrow 0$. $\square$

Lemma C.2 (Moment aggregation). *Suppose $a_i \leq b_i$, $b_i^2 \leq C(P/F)K_i$, $\sum_i K_i = P$, and $\sum_i \sqrt{K_i}a_i \leq C\sqrt{PJ}$. Then*

$$\sum_i \frac{b_i^2}{1 + \log(b_i/a_i)} \leq \frac{C(P^2/F)}{1 + \log_+(P/\sqrt{FJ})}.$$

Indices with $K_i = 0$ contribute zero.

Proof. Use (43). For $0 \leq s \leq 1$,

$$\begin{aligned} \sum_i b_i^{2-s} a_i^s &\leq C(P/F)^{1-s/2} \sum_i K_i^{1-s} (\sqrt{K_i} a_i)^s \\ &\leq C(P/F)^{1-s/2} P^{1-s} \left(\sum_i \sqrt{K_i} a_i \right)^s, \end{aligned}$$

where the second line is Hölder. This is $C(P^2/F)(\sqrt{FJ}/P)^s$. For $s \geq 1$, use $(a_i/b_i)^s \leq a_i/b_i$. Integration gives the claim when $P \geq \sqrt{FJ}$; otherwise the trivial $s = 0$ estimate gives the same statement with denominator one. $\square$

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